

Zemimet SR 50/500 mg film-coated tablets
Zemimet SR 50/1000 mg film-coated tablets
(Gemigliptin Tartrate Sesquihydrate/Metformin Hydrochloride)

1. NAME OF THE MEDICINAL PRODUCT

Zemimet SR 50/500 mg film-coated tablets

Zemimet SR 50/1000 mg film-coated tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Zemimet SR 50/500 mg film-coated tablets

Each tablet of Zemimet SR 50/500mg contains gemigliptin tartrate sesquihydrate, equivalent to 50 mg gemigliptin, and 500 mg of metformin hydrochloride.

Zemimet SR 50/1000 mg film-coated tablets

Each tablet of Zemimet SR 50/1000mg contains gemigliptin tartrate sesquihydrate, equivalent to 50 mg gemigliptin, and 1000 mg of metformin hydrochloride.

3. PHARMACEUTICAL FORM

Zemimet SR 50/500 mg film-coated tablets

Zemimet SR 50/500mg is oval-shaped orange colored film-coated tablet, debossed with “LG” symbol and a wave pattern on one side and with “5050” on the other.

Zemimet SR 50/1000 mg film-coated tablets

Zemimet SR 50/1000mg is Oblong-shaped brown color, film-coated tablet, debossed with ‘LG’ symbol and wave pattern on the one side and is engraved with ‘5010’ on the other.

4. CLINICAL PARTICULARS

4.1 Therapeutic indication

Zemimet SR is indicated as an adjunct to diet and exercise to improve glycemic control in adults with type 2 diabetes mellitus.

Zemimet SR can be administered

1. As initial therapy for treatment naive patients with inadequate glycemic control by diet and exercise.
2. In patients already receiving an identical combination of gemigliptin and metformin as separate tablets.
3. In patients with inadequate glycemic control with the maximal tolerated dose of metformin monotherapy.
4. In combination with sulfonylurea in patients with inadequate glycemic control with sulfonylurea and metformin dual therapy.
5. In combination with insulin in patients with inadequate glycemic control with insulin monotherapy or insulin and metformin dual therapy.
6. In combination with dapagliflozin in patients with inadequate glycemic control with dapagliflozin and metformin dual therapy.

4.2 Posology and method of administration

Posology (Refer Section 5.1: Clinical Efficacy and Safety)

Dosage of Zemimet SR must be individualized on the basis of the patient's current regimen, effectiveness, and tolerability. The dosage should be gradually escalated to reduce the gastrointestinal side effects associated with the metformin component.

Zemimet SR should generally be administered once daily with a meal in the evening.

- The recommended starting dose of Zemimet SR in patients who need 50 mg of gemigliptin and who are not currently treated with metformin is 50 mg gemigliptin and 500 mg metformin extended-release once daily with gradual dose escalation to reduce the gastrointestinal side effects due to metformin.
- For patients with inadequate glycemic control on metformin monotherapy or for patients with inadequate glycemic control on dual combination of metformin and sulfonylurea, or for patients with inadequate glycemic control on dual combination of metformin and insulin, or for patients with inadequate glycemic control on dual combination of metformin and dapagliflozin, the recommended daily dose of Zemimet SR is 50 mg gemigliptin and the dose of metformin already being taken.
- When used in combination with a sulfonylurea or insulin, a lower dose of the sulfonylurea or insulin may be required to reduce the risk of hypoglycemia.
- For patients switching from co-administration of gemigliptin and metformin, Zemimet SR should be initiated at the dose of gemigliptin and metformin already being taken.
- When switching from metformin immediate release to extended release, glycemic control should be closely monitored and the dose adjusted on the basis of blood glucose measurements.

No studies have been performed specifically examining the safety and efficacy of Zemimet SR in patients previously treated with other oral anti-hyperglycemic agents and switched to Zemimet SR. Any change in therapy of type 2 diabetes should be undertaken with care and appropriate monitoring as changes in glycemic control can occur.

The maximum recommended daily dose is 50 mg gemigliptin and 2,000 mg metformin extended-release.

Special populations

Renal impairment

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

The maximum daily dose of metformin should preferably be divided into 2-3 daily doses. Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin in patients with GFR<60 ml/min. If no adequate strength of Zemimet SR is available, individual monocomponents should be used instead of the fixed dose combination.

GFR (mL/min)	Metformin	Gemigliptin
60-89	Maximum daily dose is 2,000 mg. Dose reduction may be considered in relation to declining renal function.	No dose adjustment
45-59	Maximum daily dose is 2,000 mg.	No dose adjustment

	The starting dose is at most half of the maximum dose.	
30-44	Maximum daily dose is 1,000 mg. The starting dose is at most half of the maximum dose.	No dose adjustment
<30	Metformin is contraindicated	No dose adjustment

Hepatic Impairment

Zemimet SR should not be used in patients with hepatic impairment.

Elderly

As metformin and gemigliptin are excreted by the kidney, Zemimet SR should be used with caution as age increases. Monitoring of renal function is necessary to aid in prevention of metformin-associated lactic acidosis, particularly in the elderly). Clinical experience with patients ≥ 65 years of age is limited and caution should be exercised when treating this population.

Cardiac Impairment

There is limited clinical experience in patients with New York Heart Association (NYHA) Class I-II cardiac status. Therefore, gemigliptin should be used with caution in this population. Zemimet SR is not recommended in patients with NYHA Class III - IV cardiac status.

Pediatric Population

Safety and effectiveness in children and adolescents below 18 years of age have not been established. No data are available.

Method of Administration***Oral***

Zemimet SR should generally be given once daily with a dinner meal to reduce the gastrointestinal adverse reactions associated with metformin.

Zemimet SR should be swallowed as a whole and it must not be split, crushed, or chewed before swallowing.

If a dose is missed, it should be taken as soon as the patient remembers. A double dose should not be taken on the same day.

4.3 Contraindications

Zemimet SR is contraindicated in patients with:

- hypersensitivity to the active substances or to any of the excipients listed at the end of this leaflet, or a history of serious hypersensitivity reactions, i.e., angioedema or anaphylaxis, to another dipeptidyl peptidase-4 (DPP4) inhibitor;
- patients with congestive heart failure requiring medication.
- any type of acute or chronic metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)
- diabetic pre-coma;
- known hypersensitivity to components of this drug or biguanides.
- severe renal failure (GFR < 30 mL/min);
- acute conditions with the potential to alter renal function such as:
 - dehydration,
 - severe infection,
 - shock,
 - intravascular administration of iodinated contrast agents;

- any surgical procedures
- acute or chronic disease which may cause tissue hypoxia such as:
 - cardiac or respiratory failure,
 - recent myocardial infarction,
 - shock;
- debilitated, malnourished or starved patients, and those with adrenal or pituitary insufficiency.
- hepatic impairment;
- acute alcohol intoxication, alcoholism;
- pulmonary infarction, severe respiratory insufficiency, and any conditions that induce hypoxemia, gastrointestinal symptoms such as diarrhea and vomiting.
- pregnant or potentially pregnant women and breast-feeding women.

4.4 Special warnings and precautions for use

General

Zemimet SR is not a substitute for insulin in insulin-requiring patients and should not be used in patients with type 1 diabetes

Hypersensitive reaction

As Zemimet SR contains gemigliptin and metformin, it should not be used in patients who have had any serious hypersensitivity reaction to a dipeptidyl peptidase 4 (DPP4) inhibitor or a biguanide.

Acute pancreatitis

Pancreatitis has been reported in patients taking DPP-4 inhibitors including gemigliptin. Therefore, patients should be informed of the characteristic symptoms of acute pancreatitis: persistent, severe abdominal pain. If pancreatitis is suspected, gemigliptin should be discontinued and should not be restarted. Caution should be exercised in patients with a history of pancreatitis.

Use with medicinal products known to cause hypoglycemia

Sulfonylurea or insulin is known to cause hypoglycemia. Therefore, a lower dose of sulfonylurea may be required to reduce the risk of hypoglycemia when used in combination with Zemimet SR.

Severe and disabling arthralgia

There have been post-marketing reports of severe and disabling arthralgia in patients taking other DPP-4 inhibitors. The time to onset of symptoms following initiation of drug therapy varied from one day to years. Patients experienced relief of symptoms upon discontinuation of the medication. A subset of patients experienced a recurrence of symptoms when restarting the same drug or a different DPP-4 inhibitor. Consider DPP-4 inhibitors as a possible cause for severe joint pain and discontinue drug if appropriate.

Lactic acidosis

Lactic acidosis, a rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe vomiting, diarrhea, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterized by acidotic dyspnea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (< 7.35), increased plasma lactate levels (> 5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Diagnosis

Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain and hypothermia followed by coma. Diagnostic laboratory findings are decreased blood pH, plasma lactate levels above 5 mmol/l, and an increased anion gap and lactate/pyruvate ratio. If metabolic acidosis is suspected, treatment with the medicinal product should be discontinued and the patient hospitalised immediately.

Renal function

Metformin and gemigliptin are excreted by the kidney. Metformin-related lactic acidosis increases with the degree of impairment of renal function; serum creatinine concentrations should be determined regularly:

- at least once a year in patients with normal renal function
- at least two to four times a year in patients with serum creatinine levels at or above the upper limit of normal and in elderly patients.

GFR should be assessed before treatment initiation and regularly thereafter. Zemimet SR is contraindicated in patients with $\text{GFR} < 30$ mL/min and should be temporarily discontinued in the presence of conditions that alter renal function.

Decreased renal function in elderly patients is frequent and asymptomatic. Special caution should be exercised in situations where renal function may become impaired, for example when initiating anti-hypertensive or diuretic therapy or when starting treatment with a nonsteroidal anti-inflammatory drug (NSAID).

Bullous pemphigoid

There have been post-marketing reports of bullous pemphigoid requiring hospitalization in patients taking other DPP-4 inhibitors. In reported cases, patients typically recovered with topical or systemic immunosuppressive treatment and discontinuation of the DPP-4 inhibitor. Patient should be instructed to report development of blisters or erosions to their doctors while receiving Zemimet SR. If bullous pemphigoid is suspected, Zemimet SR should be discontinued and a dermatologist should be consulted for diagnosis and proper treatment.

Surgery

As Zemimet SR contains metformin, the treatment should be discontinued 48 hours before elective surgery with general, spinal or epidural anaesthesia. Zemimet SR should not usually be resumed earlier than 48 hours afterwards and only after renal function has been re-evaluated

and found to be normal.

Administration of iodinated contrast agent

The intravascular administration of iodinated contrast agents in radiological studies can lead to renal failure which has been associated with lactic acidosis in patients receiving metformin. Therefore, Zemimet SR must be discontinued prior to, or at the time of the test and not reinstituted until 48 hours afterwards, and only after renal function has been re-evaluated and found to be normal.

Regular monitoring of thyroid-stimulating hormone (TSH) levels is recommended in patients with hypothyroidism.

Long-term treatment with metformin has been associated with a decrease in vitamin B12 serum levels which may cause peripheral neuropathy. Monitoring of the vitamin B12 level is recommended.

Other precautions:

- All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.
- The usual laboratory tests for diabetes monitoring should be performed regularly.
- The tablet shells may be present in the feces. Patients should be advised that this is normal.

4.5 Interaction with other medicinal products and other forms of interaction

Co-administration of multiple doses of gemigliptin and metformin did not meaningfully alter the pharmacokinetics of either gemigliptin or metformin in patients with type 2 diabetes.

There have been no formal interaction studies for Zemimet SR. The following statements reflect the information available on the individual active substances.

Gemigliptin

In vitro assessment of interactions

The responsible enzyme for the metabolism of gemigliptin is CYP3A4. *In vitro* studies indicated that gemigliptin and its active metabolite are not inhibitors of CYP1A2, 2A6, 2B6, 2C9, 2C19, 2D6, 2E1 or 3A4 and are not inducers of CYP1A2, 2C8, 2C9, 2C19, or 3A4. Therefore, gemigliptin is unlikely to cause interactions with other drugs that utilize these metabolic pathways. *In vitro* studies further indicated that gemigliptin did not induce p-glycoprotein (P-gp) while mildly inhibited P-gp mediated transport at high concentration. Therefore, gemigliptin is unlikely to cause interactions with other P-gp substrates at therapeutic concentrations.

In vivo assessment of interactions

Effects of gemigliptin on other medicinal products

In clinical studies, gemigliptin did not meaningfully alter the pharmacokinetics of metformin, pioglitazone and glimepiride, rosuvastatin, dapagliflozin, or empagliflozin providing *in vivo* evidence of a low propensity for causing drug interactions with substrates of CYP2C8, CYP2C9, CYP3A4, organic cation transporter (OCT), and UDP-glucuronosyltransferase (UGT).

Metformin: Repeated co-administration of 50 mg gemigliptin with 2000 mg metformin, a substrate of OCT1 and OCT2, decreased the C_{max} of metformin by 13% but did not affect the AUC of metformin at steady state.

Pioglitazone: Repeated co-administration of 200 mg gemigliptin with 30 mg pioglitazone, a substrate of CYP2C8 and 3A4, decreased the AUC and $C_{\max}$ of pioglitazone by 15% and 17%, respectively. However, those of the active metabolites of pioglitazone were not changed at steady state.

Glimepiride: Co-administration of multiple doses of 50 mg gemigliptin with a single dose of 4 mg glimepiride, a substrate of CYP2C9, did not meaningfully alter the pharmacokinetics of glimepiride at steady state.

Rosuvastatin: Repeated co-administration of 50 mg gemigliptin with 20 mg rosuvastatin, a substrate of CYP2C9 and 3A4, did not meaningfully alter the pharmacokinetics of rosuvastatin at steady state.

Dapagliflozin: Repeated co-administration of 50 mg gemigliptin with 10mg of dapagliflozin, substrate of UGT1A9, did not meaningfully alter the pharmacokinetics of dapagliflozin at steady state.

Empagliflozin: Repeated co-administration of 50 mg gemigliptin with 25mg of empagliflozin, substrate of UGT2B7, UGT1A3, UGT1A8, and UGT1A9, did not meaningfully alter the pharmacokinetics of empagliflozin at steady state.

Effects of other medical products on gemigliptin

In clinical studies, metformin and pioglitazone did not meaningfully alter the pharmacokinetics of gemigliptin. Ketoconazole did not meaningfully alter the pharmacokinetics of gemigliptin and its active metabolite. Therefore, strong and moderate CYP3A4 inhibitors would not cause clinically meaningful drug interactions. Rifampicin (rifampin), on the other hand, significantly decreased exposure of gemigliptin. Therefore, co-administration with other strong CYP3A4 inducers, including rifampicin (rifampin), dexamethasone, phenytoin, carbamazepine, rifabutin and phenobarbital, is not recommended.

Metformin: Repeated co-administration of 50 mg gemigliptin with 2000 mg metformin, a substrate of OCT1 and OCT2, did not meaningfully alter the pharmacokinetics of gemigliptin and its active metabolite at steady state.

Pioglitazone: Repeated co-administration of 200 mg gemigliptin with 30 mg of pioglitazone, a substrate of CYP2C8 and 3A4, did not meaningfully alter the pharmacokinetics of gemigliptin and its active metabolite at steady state.

Ketoconazole: Co-administration of multiple doses of 400 mg ketoconazole, a strong inhibitor of CYP3A4, with a single dose of 50 mg gemigliptin increased the AUC of active moiety, the sum of gemigliptin and its active metabolite, by 1.9-fold at steady state.

Rifampicin: Co-administration of multiple doses of 600 mg rifampicin, a strong inducer of CYP3A4, with a single dose of 50 mg gemigliptin decreased the AUC and $C_{\max}$ of gemigliptin by 80% and 59%, respectively. The $C_{\max}$ of active metabolite of gemigliptin was not significantly affected while the AUC was decreased by 36% at steady state.

Rosuvastatin: Repeated co-administration of 50 mg gemigliptin with 20 mg rosuvastatin did not meaningfully alter the pharmacokinetics of gemigliptin at steady state.

Dapagliflozin: Repeated co-administration of 50mg gemigliptin with 10mg of dapagliflozin did not meaningfully alter the pharmacokinetics of gemigliptin at steady state.

Empagliflozin: Repeated co-administration of 50mg gemigliptin with 25mg of empagliflozin

did not meaningfully alter the pharmacokinetics of gemigliptin at steady state.

Metformin

Combinations not recommended

Alcohol

There is increased risk of lactic acidosis in acute alcohol intoxication (particularly in the case of fasting, malnutrition or hepatic impairment) due to the metformin active substance of Zemimet SR. Consumption of alcohol and medicinal products containing alcohol should be avoided.

Iodinated contrast agents

The intravascular administration of iodinated contrast agents in radiological studies may lead to renal failure, resulting in metformin accumulation and a risk of lactic acidosis. Therefore Zemimet SR must be discontinued prior to, or at the time of the test and not reinstituted until 48 hours afterwards, and only after renal function has been re-evaluated and found to be normal.

Combination requiring precautions for use

Some medicinal products can adversely affect renal function which may increase the risk of lactic acidosis, e.g. NSAIDs, including selective cyclo-oxygenase (COX) II inhibitors, ACE inhibitors, angiotensin II receptor antagonists and diuretics, especially loop diuretics. When starting or using such products in combination with metformin, close monitoring of renal function is necessary.

Glucocorticoids

Glucocorticoids (given by systemic and local routes), beta-2agonists, and diuretics have intrinsic hyperglycemic activity. The patient should be informed and more frequent blood glucose monitoring performed, especially at the beginning of treatment with such medicinal products. If necessary, the dose of the anti-hyperglycemic medicinal product should be adjusted during therapy with the other medicinal product and on its discontinuation.

ACE-inhibitors

ACE-inhibitors may decrease the blood glucose levels. If necessary, the dose of the anti-hyperglycemic medicinal product should be adjusted during therapy with the other medicinal product and on its discontinuation.

Phenprocoumon

Metformin may decrease the anti-coagulant effect of phenprocoumon. Therefore, a close monitoring of the INR is recommended.

Levothyroxine

Levothyroxine can reduce the hypoglycemic effect of metformin. Monitoring of blood glucose levels is recommended, especially when thyroid hormone therapy is initiated or stopped, and the dosage of metformin must be adjusted if necessary.

Organic cation transporters (OCT)

Metformin is a substrate of both transporters OCT1 and OCT2.

Co-administration of metformin with

- Inhibitors of OCT1 (such as verapamil) may reduce efficacy of metformin.
- Inducers of OCT1 (such as rifampicin) may increase gastrointestinal absorption and efficacy of metformin.

- Inhibitors of OCT2 (such as cimetidine, dolutegravir, ranolazine, trimethoprim, vandetanib, isavuconazole) may decrease the renal elimination of metformin and thus lead to an increase in metformin plasma concentration.

- Inhibitors of both OCT1 and OCT2 (such as crizotinib, olaparib) may alter efficacy and renal elimination of metformin.

Caution is therefore advised, especially in patients with renal impairment, when these drugs are coadministered with metformin, as metformin plasma concentration may increase. If needed, dose adjustment of metformin may be considered as OCT inhibitors/inducers may alter the efficacy of metformin.

4.6 Fertility, pregnancy and lactation

Pregnancy

No studies have been conducted with the combined active substances of Zemimet SR. There are no adequate and well-controlled studies in pregnant women with gemigliptin. Animal studies have shown reproductive toxicity at high doses of gemigliptin. Because of the high safety margin, no relevant risk is suggested for pregnant women.

To date, no relevant epidemiological data are available with regard to the use of metformin. Animal studies with metformin do not indicate harmful effects with respect to pregnancy, embryonic or foetal development, parturition or postnatal development.

Zemimet SR should not be used during pregnancy. If a patient wishes to become pregnant or if a pregnancy occurs, treatment with Zemimet SR should be discontinued and switched to insulin treatment as soon as possible.

Lactation

No studies have been conducted with the combined active substances of Zemimet SR. There is no information on excretion of gemigliptin or metformin into human milk. In animal studies performed with the individual active substances, both gemigliptin and metformin are excreted in the milk of lactating rats.

Zemimet SR must therefore not be used in women who are breastfeeding.

Fertility

No studies have been conducted with the combined active substances of Zemimet SR.

No studies on the effect on human fertility have been conducted for gemigliptin.

Animal studies with gemigliptin or metformin do not indicate harmful effects to fertility in male and female rat.

4.7 Effects on ability to drive and use machines

When driving or using machines, caution should be exercised as dizziness has been reported. In addition, patients should be alerted to the risk of hypoglycemia when Zemimet SR is used in combination with other anti-diabetic medicinal products known to cause hypoglycemia (e.g. sulfonylureas, insulin).

4.8 Undesirable effects

There have been no therapeutic clinical trials conducted with Zemimet SR tablets, however bioequivalence of Zemimet SR with co-administered gemigliptin and metformin has been demonstrated.

Zemimet SRPost-marketing surveillance

Table 1 presents adverse reactions which have been reported during post-marketing surveillance. The adverse reactions are listed by SOC and PT with frequency.

Frequencies are defined as Very common ($\geq 1/10$), Common ($\geq 1/100$ to $< 1/10$), Uncommon ($\geq 1/1,000$ to $< 1/100$), Rare ($\geq 1/10,000$ to $< 1/1,000$), or not known (cannot be estimated from the available data).

Table 1. Frequency of adverse reactions reported from post-marketing surveillance of Zemimet SR

System Organ Class Adverse Reaction	Frequency N=624
Gastrointestinal disorders	
Dyspepsia	Uncommon
Nausea	Uncommon
Gastritis	Uncommon
General disorders and administration site conditions	
Chest pain	Uncommon

Spontaneous global post-marketing reports

The following adverse reactions have been additionally reported during post-marketing use of the product. Because these events are reported voluntarily from a population of uncertain size, it is generally not possible to reliably estimate their frequency or establish a causal relationship to drug exposure.

• Anaemia, Palpitations, Tinnitus, Vertigo, Eye pain, Macular oedema, Abdominal discomfort, Abdominal distension, Abdominal pain, Constipation, Diarrhoea, Dry mouth, Vomiting, Asthenia, Chills, Discomfort, Drug ineffective, Face oedema, Fatigue, Oedema, Oedema peripheral, Pain, Treatment noncompliance, Hypersensitivity, Pyelitis, Rhinitis, Vaginal infection

Gemigliptin

There were 2,281 patients with type 2 diabetes, including 1,351 patients that were randomized into gemigliptin group in 10 controlled clinical safety and efficacy studies conducted to evaluate the effects of gemigliptin on glycemic control.

Safety data were collected from a total of 1,606 patients exposed to gemigliptin at a dose of 50 mg per day in controlled clinical trials of monotherapy or combination therapy of 12 weeks or more. Of these patients, 221 patients received gemigliptin alone and 1,385 received it in combination with other medications.

Serious adverse reactions related to gemigliptin have not been reported in all 10 clinical studies. Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice.

Tabulated summary of adverse reactions

Table 2 presents adverse reactions for over 12 weeks which have been reported from the pooled analysis of 10 randomized, controlled clinical studies. The adverse reactions are listed by SOC (system organ class) and PT (Preferred Term) with frequency.

Frequencies are defined as Very common ($\geq 1/10$), Common ($\geq 1/100$ to $< 1/10$), Uncommon ($\geq 1/1,000$ to $< 1/100$), Rare ($\geq 1/10,000$ to $< 1/1,000$), or not known (cannot be estimated from the available data).

Table 2. Frequency of adverse reactions for 12 or more by system organ class in patients treated with gemigliptin from the pooled analysis

System Organ Class	Number of Patients with Adverse Drug Reactions (percentage)	Frequency
Adverse		
Gastrointestinal disorders		
Dyspepsia	12 (0.7)	Uncommon
Constipation	4 (0.2)	Uncommon
Diarrhea	4 (0.2)	Uncommon
Nausea	4 (0.2)	Uncommon
Abdominal pain upper	3 (0.2)	Uncommon
Gastric disorder	2 (0.1)	Uncommon
Gastrointestinal disorder	2 (0.1)	Uncommon
Eructation	1 (< 0.1)	Rare
Gastritis hemorrhagic	1 (< 0.1)	Rare
Hyperchlorhydria	1 (< 0.1)	Rare
General disorders and administration site conditions		
Swelling	1 (< 0.1)	Rare
Infections and infestations		
Asymptomatic bacteriuria	4 (0.2)	Uncommon
Nasopharyngitis	3 (0.2)	Uncommon
Upper respiratory tract infection	1 (< 0.1)	Rare
Urinary tract infection	1 (< 0.1)	Rare
Investigations		
Lipase increased	10 (0.6)	Uncommon
Amylase increased	3 (0.2)	Uncommon
Blood creatine phosphokinase	2 (0.1)	Uncommon
Alanine aminotransferase abnormal	1 (< 0.1)	Rare
Alanine aminotransferase increase	1 (< 0.1)	Rare
Hepatic enzyme increased	1 (< 0.1)	Rare
Pancreatic enzymes increased	1 (< 0.1)	Rare
Weight decreased	1 (< 0.1)	Rare
Weight increased	1 (< 0.1)	Rare
Metabolism and nutrition disorders		
Hypoglycemia	17 (1.1)	Common
Decreased appetite	3 (0.2)	Uncommon
Hyperglycemia	2 (0.1)	Uncommon
Hypoglycemia unawareness	2 (0.1)	Uncommon
Hypertriglyceridemia	1 (< 0.1)	Rare

Musculoskeletal and connective tissue disorders		
Arthralgia	1 (< 0.1)	Rare
Myalgia	1 (< 0.1)	Rare
Nervous system disorders		
Dizziness	2 (0.1)	Uncommon
Headache	2 (0.1)	Uncommon
Diabetic neuropathy	1 (< 0.1)	Rare
Hypoaesthesia	1 (< 0.1)	Rare
Tension headache	1 (< 0.1)	Rare
Psychiatric disorders		
Insomnia	1 (< 0.1)	Rare
Reproductive system and breast disorders		
Vulvovaginal pruritus	1 (< 0.1)	Rare
Respiratory, thoracic and mediastinal disorders		
Epistaxis	1 (< 0.1)	Rare
Skin and subcutaneous tissue disorders		
Dermatitis	1 (< 0.1)	Rare
Photosensitivity reaction	1 (< 0.1)	Rare
Pruritis	1 (< 0.1)	Rare
Rash	1 (< 0.1)	Rare
Urticaria	1 (< 0.1)	Rare

Post-marketing surveillance

Table 3 presents adverse reactions which have been reported during post-marketing surveillance. The adverse reactions are listed by SOC and PT with frequency.

Frequencies are defined as Very common ($\geq 1/10$), Common ($\geq 1/100$ to $< 1/10$), Uncommon ($\geq 1/1,000$ to $< 1/100$), Rare ($\geq 1/10,000$ to $< 1/1,000$), or not known (cannot be estimated from the available data).

Table 3. Frequency of adverse reactions reported from post-marketing surveillance of Zemiglo

System Organ Class Adverse Reaction	Frequency N=3,036
Gastrointestinal disorders	
Dyspepsia	Uncommon
Constipation	Rare
Nausea	Rare
Nervous system disorders	
Headache	Rare
Dizziness	Rare
Skin and subcutaneous tissue disorders	
Rash	Rare
Metabolism and nutrition disorders	

Hypoglycemia	Rare
Investigations	
Blood creatinine increased	Rare
Ear and labyrinth disorders	
Motion sickness	Rare

Spontaneous global post-marketing reports

The following adverse reactions have been additionally reported during post-marketing use of the product. Because these events are reported voluntarily from a population of uncertain size, it is generally not possible to reliably estimate their frequency or establish a causal relationship to drug exposure.

•Pneumonia, Vomiting, Edema peripheral, Asthenia, Stevens-Johnson syndrome, Rash pruritic, Pyrexia, Gastritis, Arthralgia, Swelling face, Lip swelling, Pancreatitis

Description of selected adverse reactions

Hypoglycemia

Hypoglycemia was commonly reported in controlled clinical trials of gemigliptin over 12 weeks. In all studies, the majority of reported hypoglycemia was mild and recovered. The incidence of hypoglycemia was similar among treatment groups, including placebo, in all studies. The incidence of hypoglycemia was higher in the sulfonylurea and insulin add-on studies compared to the other studies.

Pancreatitis

Although pancreatitis has not been reported, increased levels of lipase, for which a causal relationship with gemigliptin cannot be ruled out, have been reported, though uncommonly. In the results of controlled clinical trials of monotherapy and combination therapy over 12 weeks with gemigliptin, increased lipase levels were observed in 0.6% of patients, which were mild or moderate, and most recovered.

Hypersensitivity

Hypersensitivity reactions related to gemigliptin have not been reported in all 10 clinical studies.

Bullous pemphigoid

There have been post-marketing reports of bullous pemphigoid requiring hospitalization in patients taking other DPP-4 inhibitors. There have been no reports of bullous pemphigoid in monotherapy and combination therapy controlled clinical trials of gemigliptin of 12 weeks or longer.

4.9 Overdose

No data are available with regard to overdose of Zemimet SR.

Gemigliptin

During clinical trials in healthy subjects, multiple doses of up to 600 mg gemigliptin were administered for duration of 10 days. One case of increased heartbeat was observed at a single dose of 600 mg gemigliptin. There is no experience with daily doses above 600 mg in clinical studies. In the event of an overdose, it is reasonable to employ the usual supportive measures, e.g., remove unabsorbed material from the gastrointestinal tract, employ clinical monitoring (including obtaining an electrocardiogram), and institute supportive therapy as indicated by the

patient's clinical status.

Metformin

Hypoglycemia has not been seen with metformin doses of up to 85 g, although lactic acidosis has occurred in such circumstances. High overdose or concomitant risks of metformin may lead to lactic acidosis. The most effective method to remove lactate and metformin is haemodialysis. Pancreatitis may occur in the context of a metformin overdose.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Drugs used in diabetes, Combinations of oral blood glucose lowering drugs, ATC code: A10BD18.

Mechanism of action and pharmacodynamic effects

Zemimet SR combines two anti-hyperglycemic medicinal products with complementary mechanisms of action to improve glycemic control in patients with type 2 diabetes: gemigliptin tartrate sesquihydrate, a dipeptidyl peptidase 4 (DPP-4) inhibitor, and metformin hydrochloride, a member of the biguanide class.

Gemigliptin

Mechanism of Action

Gemigliptin is a member of a class of oral anti-hyperglycemic agents called dipeptidyl peptidase 4 (DPP-4) inhibitors, which enhances the level of active incretin hormones, including GLP-1 and GIP, thereby reducing blood glucose levels. Active GLP-1 and GIP promote insulin production and release from pancreatic beta cells. GLP-1 also lowers the secretion of glucagon from pancreatic alpha cells, thereby resulting in a decreased hepatic glucose production. However these incretins are rapidly degraded by the DPP-4. Gemigliptin selectively inhibits DPP-4 activity, enhancing prolonged activation of incretin hormones. Gemigliptin demonstrates > 3,400-fold and > 9,500-fold selectivity versus DPP-9 and DPP-8, respectively.

Clinical Efficacy and Safety

Over 2,200 patients with type 2 diabetes have been included in randomized, controlled clinical trials. Overall, gemigliptin improved glycemic control when used as monotherapy or in combination treatment.

Gemigliptin dose finding

The efficacy and safety of gemigliptin monotherapy was evaluated in a placebo-controlled Phase II study of 12 week duration. The mean change in HbA1c from baseline at Week 12 was -0.98%, -0.74% and -0.78% (when adjusted with placebo data, -0.92%, -0.68% and -0.72%) at dosage levels of 50 mg, 100 mg and 200 mg, respectively.

Gemigliptin as monotherapy

The efficacy and safety of gemigliptin monotherapy was evaluated in a multinational, multicenter, randomized, placebo-controlled, parallel group, double-blind, 24-week study. The mean HbA1c change from baseline at Week 24 was -0.64% for this drug and 0.08% for placebo. The difference in HbA1c change compared to placebo was -0.71% (95% CI: -1.04, -0.37), which was a statistically significant decrease (p-value<0.0001).

Gemigliptin as add-on to metformin therapy

A multicenter, multinational, randomized, active-controlled, parallel group, double-blind, 24-week trial was conducted to evaluate the efficacy and safety of gemigliptin 50 mg compared with sitagliptin 100 mg added to ongoing metformin therapy in patients with type 2 diabetes inadequately controlled with metformin ($\geq 1,000$ mg/day) alone. Gemigliptin 50 mg once daily, gemigliptin 25 mg twice daily, or sitagliptin 100 mg once daily were administered, and after 24 weeks, the mean difference in HbA1c change from baseline was -0.87% for gemigliptin 50 mg (25 mg twice daily), -0.82% for gemigliptin 50 mg, and -0.92% for sitagliptin 100 mg. The difference in HbA1c change compared to sitagliptin was 0.06% (90% CI upper limit: 0.23) (50 mg once daily) and 0.04% (90% CI upper limit: 0.20) (25 mg twice daily), demonstrating non-inferiority (non-inferiority margin: 0.4).

Gemigliptin as add-on to a combination of metformin and sulfonylurea therapy

A multicenter, randomized, placebo-controlled, parallel group, double-blind, 24-week trial was conducted to evaluate the efficacy and safety of added gemigliptin 50 mg qd in patients with type 2 diabetes inadequately controlled with a combination therapy of metformin ($\geq 1,000$ mg/day) and glimepiride (≥ 4 mg/day or maximum tolerated dose). After 24 weeks, the mean change in HbA1c from baseline was -0.73% and 0.14% in the gemigliptin and placebo groups, respectively. The difference in HbA1c change compared to placebo was -0.87% (95% CI: -1.09, -0.64), which was a statistically significant decrease (p-value < 0.0001).

Gemigliptin as add-on to insulin (+/- metformin) therapy

A multinational, multicenter, randomized, placebo-controlled, parallel group, double-blind, 24-week trial was conducted to evaluate the efficacy and safety of added gemigliptin 50 mg compared with added placebo in subjects with type 2 diabetes who have inadequate glycemic control on insulin alone or on insulin in combination with metformin ($\geq 1,000$ mg/day). After 24 weeks, the mean change in HbA1c from baseline was -0.77% and -0.08% in the gemigliptin and placebo groups, respectively. The difference in HbA1c change compared to placebo was -0.69% (95% CI: -0.94, -0.43), which was a statistically significant decrease (p-value < 0.0001).

Gemigliptin and metformin as initial therapy

The efficacy and safety of this product were evaluated in a multinational, multicenter, randomized, active-controlled, parallel-group, double-blind, 24-week trial in which gemigliptin 50 mg and metformin (≥ 500 mg) were administered once daily to patients with type 2 diabetes with poor glycemic control. The mean change in HbA1c compared to baseline after 24 weeks was -2.06% in the combination group and -1.24% and -1.47% in the gemigliptin 50 mg and metformin monotherapy group, respectively. The change in HbA1c when compared to each single agent was statistically significant (p-value < 0.0001) at -0.82% (95% CI: -1.02, -0.63) for gemigliptin monotherapy and -0.62% (95% CI: -0.82, -0.41) for metformin monotherapy.

Gemigliptin as add-on to dapagliflozin and metformin therapy

A multicenter, randomized, placebo-controlled, parallel group, double-blind, 24-week study to evaluate the efficacy and safety of added gemigliptin 50 mg or placebo once daily in patients with type 2 diabetes who have inadequate glycemic control with dapagliflozin (10 mg /day) and metformin ($\geq 1,000$ mg/day). After 24 weeks, the mean change in HbA1c from baseline

was -0.86% and -0.20% in the gemigliptin and placebo groups, respectively. The difference in HbA1c change compared to placebo was -0.66% (95% CI: -0.80, -0.52), which was a statistically significant decrease (p-value <0.0001).

Metformin

Mechanism of action

Metformin is a biguanide with anti-hyperglycemic effects, lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycemia.

Metformin may act via three mechanisms:

- by reduction of hepatic glucose production by inhibiting gluconeogenesis and glycogenolysis
- in muscle, by modestly increasing insulin sensitivity, improving peripheral glucose uptake and utilization
- by delaying intestinal glucose absorption.

Metformin stimulates intracellular glycogen synthesis by acting on glycogen synthase. Metformin increases the transport capacity of specific types of membrane glucose transporters (GLUT-1 and GLUT-4).

In humans, independently of its action on glycemia, metformin has favorable effects on lipid metabolism. This has been shown at therapeutic doses in controlled, medium-term or long-term clinical studies: metformin reduces total cholesterol, LDLc and triglyceride levels.

Clinical efficacy and safety

The prospective randomized (UKPDS) study has established the long-term benefit of intensive blood glucose control in type 2 diabetes. Analysis of the results for overweight patients treated with metformin after failure of diet alone showed:

- a significant reduction of the absolute risk of any diabetes-related complication in the metformin group (29.8 events/1,000 patient-years) versus diet alone (43.3 events/1,000 patient-years), $p=0.0023$, and versus the combined sulfonylurea and insulin monotherapy groups (40.1 events/1,000 patient-years), $p=0.0034$
- a significant reduction of the absolute risk of any diabetes-related mortality: metformin 7.5 events/1,000 patient-years, diet alone 12.7 events/1,000 patient-years, $p=0.017$
- a significant reduction of the absolute risk of overall mortality: metformin 13.5 events/1,000

patient-years versus diet alone 20.6 events/1,000 patient-years, ($p=0.011$), and versus the combined sulfonylurea and insulin monotherapy groups 18.9 events/1,000 patient-years ($p=0.021$)

- a significant reduction in the absolute risk of myocardial infarction: metformin 11 events/1,000 patient-years, diet alone 18 events/1,000 patient-years, ($p=0.01$).

5.2 Pharmacokinetic properties

Zemimet SR

A bioequivalence study in healthy subjects demonstrated that the Zemimet SR (gemigliptin/metformin hydrochloride) combination tablets are bioequivalent to co-administration of gemigliptin and metformin hydrochloride as individual tablets.

The effects of food on pharmacokinetics of Zemimet SR (gemigliptin/metformin hydrochloride) combination tablets were similar to the known food effects of gemigliptin or metformin as individual tablets.

The following statements reflect the pharmacokinetic properties of the individual active substances of Zemimet SR.

Gemigliptin

Absorption

Following a single oral administration of gemigliptin to healthy subjects, gemigliptin was rapidly absorbed, with T_{max} occurring 1 to 5 hours post-dose. At the recommended dose of 50 mg, C_{max} and AUC were 62.7 ng/mL and 743.1 ng·hr/mL, respectively. The system exposure was increased in a dose-proportional manner in the range of 50~400 mg.

Distribution

In vitro human plasma protein binding is 29% for gemigliptin and 35%~48 % for the metabolites including the major active metabolite.

Biotransformation

The responsible enzyme for the metabolism of gemigliptin is CYP3A4. In plasma, gemigliptin and the major metabolite (LC15-0636) accounted for 65%~100% and 9%~18% of the sample radioactivity. LC15-0636, a hydroxylated metabolite of gemigliptin, is pharmacologically active and two times more potent than gemigliptin.

In vitro studies indicated that gemigliptin is not an inhibitor of CYP1A2, 2A6, 2B6, 2C9, 2C19, 2D6, 2E1 or 3A4 and is not an inducer CYP1A2, 2C8, 2C9, 2C19, or 3A4. Therefore, gemigliptin is considered unlikely to cause interactions with other drugs that utilize these metabolic pathways.

Elimination

Following oral administration of [^{14}C]Gemigliptin to healthy subjects, the administered radioactivity was recovered in feces (27%) or urine (63%). The elimination half-life after oral administration is approximately 17 hr and 24 hr for gemigliptin and LC15-0636, respectively.

Renal Impairment

The influence of renal impairment on the pharmacokinetics of gemigliptin has been evaluated. In patients with mild (CrCl: 50~80 mL/min), moderate (CrCl: 30~50 mL/min), severe (CrCl: <30 mL/min) and end stage renal disease (on hemodialysis), AUC_{inf} increased 1.20-, 2.04-, 1.50- and 1.69-fold for gemigliptin and 0.91-, 2.17-, 3.07- and 2.66-fold for LC15-0636, when

compared with the normal kidney function group. Overall active moiety, the sum of gemigliptin and LC15-0636, was increased less than or approximately 2-fold in patients with moderate and severe renal impairment.

Hepatic Impairment

The influence of hepatic impairment on the pharmacokinetics of gemigliptin has been evaluated. In mild and moderate hepatic impairment, exposure to gemigliptin (AUC) after single dosing was 50% and 80% higher than in healthy subjects. Formation of LC15-0636, a metabolite of gemigliptin, was only slightly affected by mild hepatic impairment (5% to 10% lower), while in moderate hepatic impairment, formation of LC15-0636 was about 30% lower compared to healthy subjects. Urinary excretion parameters were not markedly influenced by hepatic impairment, so the decrease in total clearance of gemigliptin observed in hepatic impairment is due a decreased metabolism rate of gemigliptin. Half-lives of gemigliptin and of LC15-0636 were slightly increased in hepatic impairment.

In mild and moderate hepatic impairment, inhibition of DPP-4 was slightly decreased compared to healthy subjects (5% to 10%), however, neither the effect on AUEC nor on $E_{\max}$ of DPP-4 inhibition was statistically significant. It is expected that dose adjustment would not be required in mild and moderate hepatic impairment based on the efficacy and safety profile of gemigliptin in clinical and non-clinical studies.

Gender

No dose adjustment is necessary based on gender. The differences in $C_{\max}$ and $AUC_{\inf}$ were not clinically significant.

Race

Caucasian subjects demonstrated 28% decrease in $C_{\max}$ and 5% decrease in $AUC_{\inf}$ when compared with Korean subjects.

Geriatric

Of the total number of patients (N=1473) in Phase II and III clinical studies, 243 (16.5%) were 65 years and over. The efficacy and safety of gemigliptin were not different between young and elderly patients. However, Zemimet SR should be used with caution in elderly patient because physiological functions including liver and kidney are usually decreased in this population.

Metformin

Absorption

After an oral dose of metformin, $T_{\max}$ is reached in 2.5 h. Absolute bioavailability of a 500 mg metformin tablet is approximately 50-60 % in healthy subjects. After an oral dose, the non-absorbed fraction recovered in feces was 20-30 %.

After oral administration, metformin absorption is saturable and incomplete. It is assumed that the pharmacokinetics of metformin absorption is non-linear. At the usual metformin doses and dosing schedules, steady state plasma concentrations are reached within 24-48 h and are generally less than 1 $\mu\text{g/mL}$. In controlled clinical trials, maximum metformin plasma levels ($C_{\max}$) did not exceed 4 $\mu\text{g/mL}$, even at maximum doses.

Interaction with food

Food decreases the extent and slightly delays the absorption of metformin. Following

administration of a dose of 850 mg, a 40 % lower plasma peak concentration, a 25 % decrease in AUC and a 35 min prolongation of time to peak plasma concentration was observed. The clinical relevance of this decrease is unknown.

Distribution

Plasma protein binding is negligible. Metformin partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean V_d ranged between 63-276 L.

Biotransformation

Metformin is excreted unchanged in the urine. No metabolites have been identified in humans.

Elimination

Renal clearance of metformin is > 400 mL/min, indicating that metformin is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 h. When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged, leading to increased levels of metformin in plasma.

5.3 Preclinical safety data

No animal studies have been conducted with Zemimet SR. Potential toxicity and reversibility to the combination of gemigliptin and metformin was evaluated in rats administered co-suspended formulation.

In the oral rat single-dose study, the approximate lethal dose levels were considered to be greater than 150 mg/kg and 1500 mg/kg for gemigliptin and metformin, respectively.

In 3-month toxicity studies in which rats were treated with either metformin or gemigliptin alone, or a combination of metformin and gemigliptin, no additional toxicity was observed from the combination. The NOAEL in these studies was observed at exposures to gemigliptin of approximately 23~26 times the human exposure (50 mg) and to metformin of approximately 13~14 times the human exposure (1,000 mg).

The following data are findings in studies performed with gemigliptin or metformin individually.

Gemigliptin

A two-year carcinogenicity study was conducted in male and female rats given oral doses of gemigliptin of 50, 150, and 450 mg/kg/day. No evidence of carcinogenicity with gemigliptin was found in either male or female rats. This dose results in exposures approximately 129~170 times the human exposure at the maximum recommended daily adult human dose (MRHD) of 50 mg/day based on AUC comparisons. A 6-month carcinogenicity study has been performed in TgrasH2

transgenic mice at doses of 200, 400, and 800 mg/kg/day in males and 200, 600, 1200 mg/kg/day in females. There was no evidence of carcinogenicity with gemigliptin at a dose of 1200 mg/kg/day, approximately 87 times the human exposure at the maximum recommended daily dose.

Genotoxicity assessments in the Ames test, chromosomal aberrations test and *in vivo* micronucleus tests in mice and rats were negative.

The fertility of gemigliptin was not affected at dose of 800 mg/kg/day in rat.

Gemigliptin was not teratogenic up to 200 mg/kg/day in rats and 300 mg/kg/day in rabbits, which are respectively 83 and 153 times human exposure at the MRHD of 50 mg/day.

Gemigliptin at dose of 800 mg/kg/day in rat, approximately 264 times human exposure at the MRHD of 50 mg/day, increased the incidence of fetus cleft palate malformation, dilated renal pelvis, misshapen thymus and sternoschisis, with increasing dose.

Metformin

Preclinical data for metformin reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential, toxicity to reproduction.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Zemimet SR 50/500 mg film-coated tablets

Tablet core

- | | |
|--|-----------------------------|
| • Microcrystalline Cellulose (Type 12) | • Croscarmellose Sodium |
| • Stearyl Fumarate Sodium | • Magnesium Stearate |
| • Polyvinyl Acetate Aqueous Dispersion 30% | • Hypromellose 2208 |
| • Carboxymethylcellulose Sodium | • Opadry II 85F43172 Orange |

Film coating

- | | |
|-----------------------------------|--------------------|
| • Polyvinyl Alcohol | • Titanium Dioxide |
| • Polyethylene Glycol 3350 | • Talc |
| • Sunset Yellow FCF Aluminum Lake | • Red Iron Oxide |

Zemimet SR 50/1000 mg film-coated tablets

Tablet core

- | | |
|--|-------------------------|
| • Microcrystalline Cellulose (Type 102) | • Croscarmellose Sodium |
| • Stearyl Fumarate Sodium | • Magnesium Stearate |
| • Polyvinyl Acetate Aqueous Dispersion 30% | • Hypromellose 2208 |
| • Opadry II 85F34790 Pink | |

Film coating

- | | |
|----------------------------|--------------------|
| • Polyvinyl Alcohol | • Titanium Dioxide |
| • Polyethylene Glycol 3350 | • Talc |
| • Red Iron Oxide | |

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

36 months

6.4 Special precautions for storage

Store below 30°C in tight container.

6.5 Nature and contents of container

Zemimet SR 50/500 mg film-coated tablets

Opaque blisters (PVC/PE/PVDC and aluminium foil lidding). In boxes of 2 blisters x 14 film-coated tablets.

Zemimet SR 50/1000 mg film-coated tablets

Opaque blister (PVC/PE/PVDC and aluminium foil lidding). In boxes of 2 blisters x 14 film-coated tablets.

6.6 Special precautions for disposal and other handling

Any unused product or waste material should be disposed of in accordance with local requirements.

7. DATE OF REVISION OF THE TEXT

23 October, 2025

Manufacturer:

LG CHEM, LTD.

151, Osongsaengmyeong1-ro, Osong-eup, Heungdeok-gu, Cheongju-si,
Chungcheongbuk-do, Korea

Product Registration Holder:

PHARMANIAGA MARKETING SDN BHD (198401005734)

No.7, Lorong Keluli 1B, Kawasan Perindustrian Bukit Raja Selatan, Seksyen 7, 40000 Shah
Alam, Selangor Darul Ehsan, Malaysia